

SCIENCE MASTER X

Complete Academic Study Notes • Class 10 Science

CHAPTER 1

Chemical Reactions And Equations

(Full Chapter)

Chemical Reactions & Characteristics • Balancing Chemical Equations • Types of Chemical Reactions (Combination, Decomposition, Displacement, Double Displacement) • Oxidation & Reduction (Redox) • Corrosion & Rancidity

Chapter Details

Target Level: CBSE Class 10

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Chapter Overview & Basic Concepts

A chemical reaction is a process in which one or more substances (reactants) react to form new substances (products) with entirely different chemical properties.

SECTION 1: CHEMICAL REACTIONS AND BALANCING EQUATIONS

1.1 Characteristics of Chemical Reactions

- **Change in State:** E.g., Combustion of wax forming water vapor and carbon dioxide.
- **Change in Color:** E.g., Rusting of iron changes gray iron to reddish-brown rust.
- **Evolution of Gas:** E.g., Zinc reacting with dilute H_2SO_4 evolves H_2 gas.
- **Change in Temperature:** E.g., Reaction of quicklime with water releases a large amount of heat.
- **Formation of Precipitate:** E.g., Mixing lead nitrate and potassium iodide forms a yellow precipitate.

1.2 Chemical Equations & Law of Conservation of Mass

- **Chemical Equation:** A symbolic representation of a chemical reaction using chemical formulas of reactants and products.
- **Balanced Chemical Equation:** An equation where the total number of atoms of each element is equal on both sides.
- **Law of Conservation of Mass:** Mass can neither be created nor destroyed in a chemical reaction. Total mass of reactants must equal the total mass of products.

Important Reaction (Balancing Example):

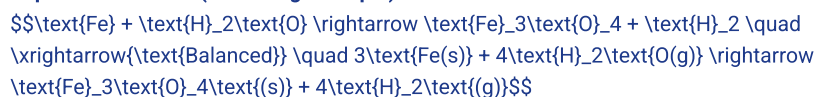


Diagram: Burning of Magnesium Ribbon in Air



NCERT High-Yield Insight: Magnesium ribbon is cleaned with sandpaper before burning to remove the protective layer of basic magnesium carbonate from its surface.

SECTION 2: TYPES OF CHEMICAL REACTIONS

2.1 Combination Reaction

- A reaction in which two or more reactants combine to form a single product.
- General Equation:** $A + B \rightarrow AB$

Important Chemical Reactions:

- Slaking of Lime (Exothermic): $\text{CaO(s)} + \text{H}_2\text{O(l)} \rightarrow \text{Ca(OH)}_2\text{(aq)} + \text{Heat}$
- Burning of Coal: $\text{C(s)} + \text{O}_2\text{(g)} \rightarrow \text{CO}_2\text{(g)}$

2.2 Decomposition Reaction

- A reaction in which a single compound breaks down into two or more simpler substances.
- Thermal Decomposition:** Carried out by heating. $2\text{FeSO}_4\text{(s [Green])} \xrightarrow{\Delta} \text{Fe}_2\text{O}_3\text{(s [Reddish Brown])} + \text{SO}_2\text{(g)} + \text{SO}_3\text{(g)}$
 $2\text{Pb(NO}_3)_2\text{(s [Colorless])} \xrightarrow{\Delta} 2\text{PbO(s [Yellow])} + 4\text{NO}_2\text{(g [Brown fumes])} + \text{O}_2\text{(g)}$
- Electrolytic Decomposition:** Carried out using electricity. $2\text{H}_2\text{O(l)} \xrightarrow{\text{Electricity}} 2\text{H}_2\text{(g [Cathode])} + \text{O}_2\text{(g [Anode])}$
- Photolytic Decomposition:** Carried out by light (used in black & white photography). $2\text{AgCl(s [White])} \xrightarrow{\text{Sunlight}} 2\text{Ag(s [Grey])} + \text{Cl}_2\text{(g)}$

2.3 Summary of Decomposition Types

Decomposition Reactions

- **Thermal:** Requires Heat Energy (Δ)
- **Electrolytic:** Requires Electric Current
- **Photolytic:** Requires Sunlight ($h\nu$)

SECTION 3: DISPLACEMENT, DOUBLE DISPLACEMENT & REDOX

3.1 Displacement & Double Displacement Reactions

Reaction Type	Definition & General Form	NCERT Example Reaction
Displacement	A more reactive element displaces a less reactive element from its salt solution. $A + BC \rightarrow AC + B$	$\text{Fe(s)} + \text{CuSO}_4\text{(aq [Blue])} \rightarrow \text{FeSO}_4\text{(aq [Light Green])} + \text{Cu(s)}$
Double Displacement	Reaction in which there is an exchange of ions between reactants to form two new compounds. $AB + CD \rightarrow AD + CB$	$\text{Na}_2\text{SO}_4\text{(aq)} + \text{BaCl}_2\text{(aq)} \rightarrow \text{BaSO}_4\text{(s [White ppt])} + 2\text{NaCl(aq)}$

3.2 Oxidation, Reduction & Redox Reactions

- Oxidation:** Gain of oxygen or loss of hydrogen.
- Reduction:** Gain of hydrogen or loss of oxygen.
- Redox Reaction:** A reaction where oxidation and reduction occur simultaneously.
- Oxidizing Agent:** Substance that gets reduced (supplies oxygen or removes hydrogen).
- Reducing Agent:** Substance that gets oxidized (supplies hydrogen or removes oxygen).

Redox Reaction Analysis:



- Substance Oxidized: $\text{H}_2 \rightarrow \text{H}_2\text{O}$
- Substance Reduced: $\text{CuO} \rightarrow \text{Cu}$
- Oxidizing Agent: CuO | Reducing Agent: H_2

Diagram: Electrolysis of Water



Cathode (H_2) | Anode (O_2)

NCERT High-Yield Insight: During the electrolysis of water, the volume of hydrogen gas collected at the cathode is double the volume of oxygen gas collected at the anode because H_2O contains hydrogen and oxygen in a 2:1 ratio by volume.

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SECTION 4: CORROSION & RANCIDITY

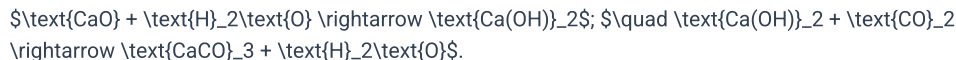
4.1 Effects of Oxidation in Daily Life

- **Corrosion:** The process in which metals are slowly eaten away by the action of air, moisture, or chemicals on their surface.
 - *Rusting of Iron:* Reddish-brown coating formed ($\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O}$).
 - *Corrosion of Copper:* Green coating formed ($\text{CuCO}_3 \cdot \text{Cu(OH)}_2$).
 - *Tarnishing of Silver:* Black coating formed (Ag_2S).
- **Rancidity:** The oxidation of fats and oils in food when exposed to air, leading to unpleasant smell and taste.
- **Prevention of Rancidity:**
 - Adding antioxidants (e.g., BHA, BHT).
 - Flushing food packages (chips) with unreactive **Nitrogen gas**.
 - Storing food in airtight containers or refrigeration.

? Important Practice Questions

Q1: Why is white washing done using quicklime? Write the reactions involved.

Ans: Quicklime (CaO) reacts with water to form slaked lime (Ca(OH)_2). When applied to walls, it reacts slowly with CO_2 in air to form a shiny white layer of calcium carbonate (CaCO_3).



Q2: Why does the color of copper sulphate solution change when an iron nail is dipped in it?

Ans: Iron is more reactive than copper, so it displaces copper from copper sulphate solution to form iron sulphate, changing the blue solution to light green.



Q3: Identify the substance oxidized and reduced in: $\text{MnO}_2 + 4\text{HCl} \rightarrow \text{MnCl}_2 + 2\text{H}_2\text{O} + \text{Cl}_2$

Ans: Substance Oxidized: HCl (to Cl_2); **Substance Reduced:** MnO_2 (to MnCl_2).

Q4: What are thermal decomposition reactions? Give one example with observation.

Ans: Reactions in which a single substance breaks down upon heating. Example: Heating lead nitrate produces brown fumes of nitrogen dioxide (NO_2).

